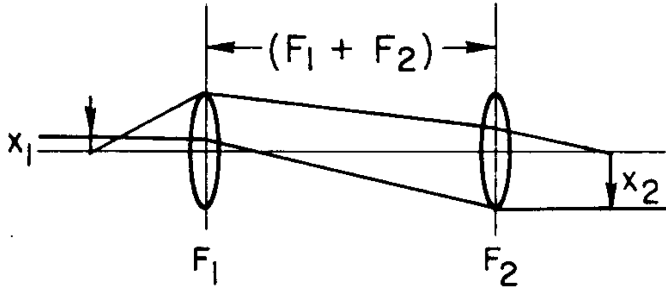


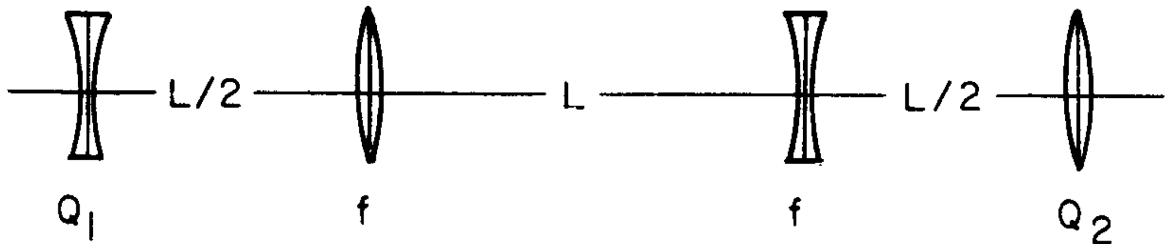
Project 2. Due Tuesday, March 3, 11:59 pm.

[Hint: check out Sec. 4.1.2–4.1.3 and Sec. 4.2.1–4.2.3 in Brown, Servranckx.]

1. A one-dimensional telescopic system consists of a drift of length f_1 , a focusing quadrupole of focal length f_1 , a drift of length $f_1 + f_2$, another focusing quadrupole of focal length f_2 , and finally another drift of length f_2 (see figure).



- Derive the transfer matrix of the system in (x, x') (show your work, not just write the final result).
 - Find the relation between β_2 and β_1 ; α_2 and α_1 .
 - Calculate the phase advance of the cell [Hint: you need both sin and cos for that.]
 - Why would this system not work in two dimensions?
 - Describe what you need to do to extend the system to two dimensions, sketch the new system layout (optional: find relationships between drift lengths and quadrupole strengths).
2. Design (on paper or using Bmad) a quarter-wave transformer that will transform your initial values of (β_1, α_1) to the final values of (β_2, α_2) . [Test it using $\beta_1 = 5$ m, $\beta_2 = 50$ m, $\alpha_1 = \alpha_2 = 0$. Once you are happy with the results (plot β_1 , β_2 , α_1 , and α_2 as a function of s), try varying α too.] Feel free to make simplifying assumptions about the combination of μ , L , f . What are the final values of L , f , Q_1 , and Q_2 ?



3. Starting with the Bmad file for the dispersion suppression section, design your own dispersion suppression section for phase advance of $\pi/3$ (as opposed to $\pi/2$).